COUNTING THE ACTIVE DEGREES OF FREEDOM OF A TRAINING RUN: WHAT A SINGLE SCALAR LOG CAN AND CANNOT MEASURE

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ABSTRACT

Within any window of training, a neural network's parameter trajectory moves on a set far smaller than the parameter space. We define the active dimension of the window as the dimension of the set the trajectory recurrently visits, which is a count of independent components rather than a score, and ask how much of it survives the projection onto a single scalar log. The estimator is standard, delay reconstruction followed by a maximum likelihood intrinsic dimension; the contribution is knowing when its output is a count. On six systems of increasing realism, ending with a linear head confined to a random subspace on a frozen nonlinear image classifier, whose component count is fixed by construction and whose excitation is verified by measurement, recovery is accurate to under one component up to eight directions on withheld ranks and seeds. Two diagnostics computed from the series alone flag the two regimes in which the value may not be read as a count. Applied to grokking they place both standard experimental settings outside the admissible regime, for two different reasons which they identify correctly, while the estimate still separates generalising from non-generalising runs in four label-matched pairs. Measured directly from the trajectory covariance, where no reconstruction is needed, the effective rank collapses by a median factor of six at the onset of generalisation in four regularised runs and not in their controls, which is the simplification the mechanistic accounts of grokking lead one to expect. Without weight decay the function instead settles early and the parameter norm never comes back down, so a low rank on its own separates nothing: what marks the transition is that the collapse is sharp and then reverses.

1 INTRODUCTION

A neural network is trained in a parameter space of dimension P, but the trajectory the optimiser follows does not fill it. Over a window it moves on a much smaller set, and the size of that set changes over a run. We call its dimension the *active dimension* of the window and define it in equation (1) as $\dim \mathcal{M}_{\mathcal{W}}$, where $\mathcal{M}_{\mathcal{W}}$ is the set of states the trajectory recurrently visits. It is a count of independent components: if the optimiser is carried by r independent quasiperiodic phases the orbit closes onto an r-torus and the active dimension is r, on a scale fixed by no free parameter. It is neither the dimension of the subspace the optimiser is confined to, which the parameterisation fixes (Li et al., 2018), nor the rank of the map from it to the outputs, nor the effective rank of the trajectory covariance, which anisotropy depresses below the count; section 3.1 separates the four. It exists only where the trajectory recurrently visits anything, and delay-embedding statistics return a plausible number whether it does or not, so most of the work below goes into deciding which case one has.

Measuring it directly requires storing the trajectory, which is expensive; a scalar log costs nothing, and a run already keeps several. We therefore ask how much of the active dimension survives the projection onto one such series. The instrument is standard, delay reconstruction (Takens, 1981; Sauer et al., 1991) followed by the pooled Levina–Bickel estimator (Levina & Bickel, 2004; MacKay & Ghahramani, 2005), and its hazard is that it returns a number when no invariant set exists: power-law noise yields a finite reproducible dimension set by its spectral exponent (Osborne & Provenzale, 1989), and a decaying transient, which is what a converging run looks like, yields a large one.

Section 4 fixes a validation protocol before any result, and two of our six systems fail one of its controls by reproducing their reported result with no training taking place.

Contributions.

1. We separate four quantities that are frequently identified with one another: the dimension of the subspace the optimiser is confined to, the rank of the map from it to the outputs, the number of independent components of the set it recurrently visits, and the effective rank of its covariance (section 3.1). Only the third is estimated here, and the fourth is measured beside it on every system so that the two cannot be conflated.
2. We evaluate the estimator on six systems of increasing realism, ending with a constrained head on a network trained on image data (section 5). It recovers the active dimension to 0.87 components up to eight directions, uncalibrated, on withheld ranks and seeds, and orders it to twenty.
3. We measure the conditions the absolute value requires (section 6), including a control that separates the effective rank from the manifold dimension, and validate two diagnostics that flag the two failure regimes from the series alone.
4. Applied to grokking (section 7), the trajectory effective rank collapses sharply at generalisation and re-expands, in four regularised runs and not in their controls, while the diagnostics place both published settings outside the regime in which a scalar log could have said so on its own.

2 Related work

Delay reconstruction. Takens' theorem (Takens, 1981), its prevalence form (Sauer et al., 1991) and its extensions to forced systems (Stark, 1999; Stark et al., 2003) license the reconstruction used here, and empirical dynamic modelling applies the same machinery to ecological and physiological series (Sugihara & May, 1990; Sugihara et al., 2012). The lag and the embedding dimension are conventionally set by mutual information (Fraser & Swinney, 1986) and false nearest neighbours (Kennel et al., 1992; Cao, 1997); the linear alternative is the singular spectrum of the delay matrix (Broomhead & King, 1986), reported throughout as $\mathrm{PR}_{\mathrm{delay}}$.

Intrinsic dimension, and when it is spurious. The dimension of the reconstructed cloud is estimated by the maximum likelihood estimator of Levina & Bickel (2004) with the pooling correction of MacKay & Ghahramani (2005), whose bias relative to the per-point average is known in closed form and is verified in section 5.1; the correlation dimension (Grassberger & Procaccia, 1983) and the neighbour-ratio estimator (Facco et al., 2017) are the standard alternatives (Camastra & Staiano, 2016). A finite record also bounds what any of them can resolve (Eckmann & Ruelle, 1992). Two spurious regimes are documented: on an oversampled record temporally adjacent neighbours make the orbit look like the smooth curve it locally is, biasing the estimate towards one (Theiler, 1986), and power-law noise produces a finite reproducible value (Osborne & Provenzale, 1989). The usual defence, a surrogate test against a linear-stochastic null (Theiler et al., 1992; Schreiber & Schmitz, 1996), answers whether nonlinear structure is present rather than how much, and misses the regime that dominates optimisation logs: a deterministic transient has neither an attractor nor noise, and returns a large stable number where the trajectory is a curve.

Dimension in optimisation. Random-subspace training (Li et al., 2018; Aghajanyan et al., 2021) fixes the available dimension and reports the smallest k at which a task is still solvable, a property of the task rather than of a window; we borrow the parameterisation and measure a different quantity inside it. The closest existing statement to the active dimension is that gradient descent occurs largely in a small subspace spanned by the top Hessian eigenvectors (Gur-Ari et al., 2018; Sagun et al., 2017), obtained from the Hessian itself, whereas the intrinsic dimension of data and of representations (Ansuini et al., 2019; Pope et al., 2021) is that of a cloud of points in data or activation space rather than of a trajectory; the participation ratio we adopt is the standard dimensionality measure for neural population activity (Gao et al., 2017), and is not the spectral-entropy quantity that shares its usual name (Roy & Vetterli, 2007). The one prior measurement of the same object we are aware of is incidental: Notsawo Jr. et al. (2023) report that two principal components carry over $98\,\%$ of the variance of a grokking trajectory. All of these need the weights, the gradients or the activations, and the nearest precedent for a scalar computed during training and then used to decide something is the gradient noise scale (McCandlish et al., 2018). Mini-batch training near a minimum is well described as a stochastic process (Mandt et al., 2017), which is the third regime of table 1; whether the oscillations of full-batch descent at the edge of stability (Cohen et al., 2021) amount to the recurrence the first regime needs is a question we raise rather than settle, and section 7.3 measures it on our own runs.

Grokking. We work in the two settings in which delayed generalisation (Power et al., 2022) is usually studied. The first is regularised and, in our version, stochastic: the transformer configuration of Nanda et al. (2023), also used by Liu et al. (2023), on modular addition and on the S_5 composition task of Power et al. (2022). The second is the unregularised full-batch gradient descent of Gromov (2023), whose closed-form solution supplies an analytic reference for the learned representation, on the modular polynomials and representability criterion of Doshi et al. (2024), which supply label-matched target pairs. Section L states where each setting departs from its source. Progress measures and explanations are defined in weight or activation space (Nanda et al., 2023; Varma et al., 2023; Liu et al., 2022; Barak et al., 2022), with Kumar et al. (2024) instead placing the transition between lazy and rich training; section 7.1 compares its result against two of them. The work closest to ours reads the loss curve alone: Notsawo Jr. et al. (2023) predict whether a run will later grok from spectral features of the loss oscillations (Thilak et al., 2022) of its early learning curve, and Merrill et al. (2023) track the sparse and dense subnetworks competing across the transition. We estimate instead a dynamical quantity with an independently measured ground truth.

3 Problem statement

3.1 Available, functional and active dimension

Let $\boldsymbol{\theta}_t \in \mathbb{R}^P$ be the parameter vector at step t, and let the optimiser be confined, by construction or by the dynamics, to an affine subspace $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{V}^\top \boldsymbol{c}$ with $\boldsymbol{V} \in \mathbb{R}^{k \times P}$ orthonormal. Over a window $\mathcal{W}$ the *available* dimension is k, fixed by the parameterisation; the *functional* dimension is the rank of $\partial \boldsymbol{f}(\boldsymbol{c})/\partial \boldsymbol{c}$ for $\boldsymbol{f}$ the outputs on a fixed probe set, bounding how many available directions can change the computed function; and the *active* dimension is

$$d_{\mathrm{act}}(\mathcal{W}) \;=\; \dim \mathcal{M}_{\mathcal{W}}, \tag{1}$$

where $\mathcal{M}_{\mathcal{W}}$ is the set of states the trajectory recurrently visits inside the window, the local part of the invariant set or the closure of the orbit. If the parameters depend on r independent quasiperiodic phases, $\boldsymbol{c}_t = F(\varphi_1(t), \ldots, \varphi_r(t))$, that closure is an r-torus and $d_{\mathrm{act}} = r$. It is not the dimension of the time-parameterised curve $t \mapsto \boldsymbol{c}_t$, which is one for every r, but of the set the curve fills, and it is what a delay reconstruction estimates. So stated it exists only where such a set does, which in two of the three regimes of section 3.3 it does not, and that is why the diagnostics of section 3.4 come before any number.

A second quantity is measured throughout and must not be confused with the first. For a non-negative spectrum $\lambda_1, \ldots, \lambda_n$ the participation ratio

$$\mathrm{PR}(\lambda) \;=\; \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2} \tag{2}$$

equals m exactly when m of the λ_i are equal and the rest vanish. Applied to $\operatorname{spec}\operatorname{Cov}\{\boldsymbol{c}_t : t \in \mathcal{W}\}$ it gives the *trajectory effective rank* $\mathrm{PR}^{\mathrm{pos}}(\mathcal{W})$, with $\mathrm{PR}^{\mathrm{upd}}$ on the increments and $\mathrm{PR}^{\mathrm{det}}$ on the positions after a per-window linear detrend, which a drifting trajectory requires. It is not a count: it equals r when r directions carry comparable variance and falls well below r when they do not, while d_{act} stays at r (section 6.4). It verifies that a construction excites the r directions it claims to, and it is the quantity measured directly in section 7.1. All four are distinct and measurably so: in section 5.3 the available dimension is 10 and the output Jacobian has rank 10, and removing the preconditioner leaves both unchanged and the injected rank intact while the effective rank falls from 7.99 to 6.43 under stochastic forcing.

Table 1: What the orbit fills in each dynamical regime, whether the active dimension of equation (1) is defined there, and what the estimator returns on the systems of section 5. Only the first regime supports reading the estimate as a count.

regime	what the orbit fills	d_{act}	estimate here
deterministic, recurrent	an r-torus	r	r
deterministic, transient	a curve, never revisited	undefined	≈ 29, at every r
stochastically driven	no low-dimensional invariant set	undefined	set by $E_{\max}$, not by r

3.2 Observation model and estimator

We observe a scalar series $x_t = \phi(\boldsymbol{s}_t)$, with $\boldsymbol{s}_t$ the full optimiser state and ϕ fixed before the run, and reconstruct $\boldsymbol{y}_t = (x_t, \dots, x_{t-(E-1)\tau}) \in \mathbb{R}^E$. For a generic pair of dynamics and observation on a compact invariant manifold of dimension d this map is an embedding once $E > 2d$ (Takens, 1981); the prevalence form replaces d by the box-counting dimension and adds conditions on periodic orbits (Sauer et al., 1991). Genericity is not verifiable for a ϕ fixed in advance, which is why every result below is reported over a family of observers.

We estimate the intrinsic dimension of $\{\boldsymbol{y}_t\}$ by maximum likelihood on nearest-neighbour distances (Levina & Bickel, 2004). With $r_1^{(i)} \leq \cdots \leq r_m^{(i)}$ the distances from point i to its m nearest neighbours, subject to a Theiler exclusion W_T on how close in time they may be (Theiler, 1986; Kantz & Schreiber, 2004), the local estimate is

$$\hat{d}_i \;=\; \left[\frac{1}{m-1}\sum_{j=1}^{m-1}\log\frac{r_m^{(i)}}{r_j^{(i)}}\right]^{-1}, \tag{3}$$

and the two ways of combining the N local estimates differ: averaging them, as Levina & Bickel (2004) propose, gives $\hat{d}_{\mathrm{LB}} = N^{-1}\sum_i \hat{d}_i$, while pooling the likelihood first, as MacKay & Ghahramani (2005) argue one should, averages the reciprocals,

$$\hat{d}_{\mathrm{MG}}^{-1} \;=\; \frac{1}{N(m-1)}\sum_{i=1}^{N}\sum_{j=1}^{m-1}\log\frac{r_m^{(i)}}{r_j^{(i)}}. \tag{4}$$

We write LB for the average of equation (3) and MG for the pooled form of equation (4). Under the local Poisson model behind both, $\hat{d}_i^{-1}$ is a sum of $m-1$ exponential variates and is unbiased where $\hat{d}_i$ is not, so LB should exceed MG by a factor $(m-1)/(m-2)$, which at the $m = 20$ used here is under six per cent. Table 6 reports both throughout. Appendix A states the algorithm as implemented.

Three further statistics accompany every MG value, since each can reproduce an apparent dimension signal with no geometry at all: the *roughness* $\mathrm{std}(\Delta x)/\,\mathrm{std}(x)$, the *linear participation ratio* $\mathrm{PR}_{\mathrm{delay}}$ of equation (2) on $\mathrm{Cov}\{\boldsymbol{y}_t\}$, and the *spectral participation ratio* of equation (2) on the power spectrum of x.

3.3 Dynamical regimes

Two conditions have to hold and they are not the same. The theorems require a deterministic flow on a compact invariant set, which is what makes $\mathcal{M}_{\mathcal{W}}$ exist. The neighbour statistic additionally requires the finite record to visit that set densely enough for near neighbours to be recurrences rather than temporal neighbours, which is what the Theiler exclusion enforces and section 3.4 tests. An optimiser near a solution may be in any of the three regimes of table 1, and only the first meets the first condition.

The third regime is what mini-batch gradient noise produces, and theory serves it worse than it looks. The embedding theorems for forced systems (Stark, 1999; Stark et al., 2003) do cover it in part: they allow the update to depend on the state arbitrarily, and their iterated-function-system form covers a driver that selects among finitely many maps, which is what sampling from a finite dataset does. What they deliver, however, is an embedding of the state space *fibre by fibre*, one fibre per realisation of the driving sequence. A single observed run crosses fibres, so its reconstructed cloud is not confined to the image of any one low-dimensional set, and there is no invariant measure on such a set for the neighbour statistic to sample. Coloured noise alone, meanwhile, yields a finite reproducible dimension fixed by its spectral exponent (Osborne & Provenzale, 1989). The second regime is equally consequential and less often noticed: a decaying transient traces a curve it never returns to, so it has no $\mathcal{M}_{\mathcal{W}}$ at all, and its effective rank measures 1.02 to 1.09 in section 5.3 only because the curve is nearly straight there.

3.4 Admissibility diagnostics

Both failure regimes can be detected without a ground truth, by two statistics and one flag. The **identifiability ratio** $\rho_{\text{ident}} = \hat{d}_{\text{MG}}(2E_{\max})/\hat{d}_{\text{MG}}(E_{\max})$ is near unity when a dimension is resolvable and grows when the estimate is a property of the embedding space rather than of the data. The **trend-crossing count** is the number of sign changes of the window's residual about its own least-squares line; it tests non-monotonicity of the series, not recurrence in the reconstructed space, and it is what separates the transient regime, where ρ_{ident} is near unity for the wrong reason. The **degeneracy flag** fires when more than one per cent of neighbour distances or per-point log-ratio sums reach their numerical floors. Reference values are in section 6.4, calibrated against no wider family, so these are flags for the failure modes tested and not a validated classifier.

4 Validation protocol

Delay-embedding statistics return a finite number on data satisfying none of their hypotheses. Six requirements are fixed in advance, and every result here is held to them.

1. *The component count is constructed and its realisation verified.* Every system is built with r independent quasiperiodic phases, so $d_{\text{act}} = r$; the effective rank is then measured on the recorded trajectory to confirm all r directions are comparably excited.
2. *The estimator configuration is selected once and frozen*, on a grid of estimator parameters only, since a grid over system parameters changes the data when one is chosen.
3. *The split is disjoint in seed and in rank*, since withholding seeds alone leaves the excitation geometry shared whenever it is a function of r.
4. *A zero learning rate must silence the observer*, which otherwise reads the external drive through the residual rather than through the optimiser.
5. *The three statistics of section 3.2 are reported on every call*, and one performing comparably is reported as such.
6. *Excitation bandwidth is matched across ranks*, or a construction in which each added direction adds a higher frequency lets the roughness alone order the rank.

5 Evaluation on systems with a known active dimension

The estimator is evaluated on six systems of increasing realism, from a prescribed oscillation with no optimiser to a constrained head on a network trained on image data. In each the orbit closes onto an r-torus by construction, the measured effective rank confirms that all r directions are excited, and the estimator sees only a scalar series (table 3).

5.1 An oscillating diagonal matrix

The first system has no optimiser and no feedback, which makes a negative result decisive and a positive one weak. Let $\boldsymbol{W}(t) = \operatorname{diag}(b_i + \delta_i(t))$ with $b_i \neq 0$ and exactly r coordinates oscillating at rationally independent frequencies inside one octave. The orbit closure is an r-torus, so $d_{\text{act}} = r$, and the amplitudes are equal, so the effective rank measures r as well. The offsets must be non-zero, since at $b_i = 0$ the squared Frobenius norm is even in each δ_i and collapses the torus onto a quotient. That removes one symmetry; it does not make a scalar observation injective, which for $r > 1$ it cannot be. What the theorems require is an injective *delay* map, which genericity supplies. One configuration was frozen on a single seed, and two further seeds and a non-monotone schedule of ranks were withheld. On those two seeds the mean absolute error over $r = 1, \ldots, 10$ is 0.30 and 0.69, but the per-rank error is not uniform: it stays within a quarter of a component to $r = 8$ and then reaches 0.82 at nine and 1.51 at ten. This is the first appearance of a ceiling that recurs in every later system.

It also supplies the control that decides whether the estimate reflects geometry at all. Driving r coordinates from one phase with distinct fixed offsets gives r visibly moving coordinates and one degree of freedom; the estimate separates that case from four independent oscillators by a factor of 3.4, where the roughness separates them by $5\,\%$. That is the clearest evidence available that the geometry does work that smoothness cannot.

5.2 Adding an optimiser, and then a network

The next four systems add an optimiser, a nonlinear link and backpropagation, and none yields a recovery number the protocol accepts. A $D = 64$ linear model and a logistic model, both fitted by full-batch gradient descent on one-hot inputs with exactly r target coordinates moving quasiperiodically, order the rank almost perfectly to twenty and saturate above about twelve directions, placing the saturation point of section 5.1 in a system with feedback and a nonlinear link; their configuration was selected in breach of requirement 2, so table 3 locates that point but measures no recovery. A trainable $\boldsymbol{z} \in \mathbb{R}^r$ optimised through a frozen nonlinear decoder, and a two-layer perceptron of $P = 243$ parameters restricted to $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{U}\boldsymbol{z}$, then add backpropagation, and read naively both succeed.

Both nevertheless fail requirement 4. With the learning rate set to zero, so that no training occurs, the perceptron experiment reproduces its reported result almost exactly, the two series correlating above 0.98 at every rank, because the drive acts on the targets and so reaches the observers through the residual without passing through the optimiser. Section 5.3 rebuilds the design with the drive on the loss weights instead.

5.3 A constrained head on a network trained on image data

The sixth system is the parameterisation of section 5.2 on real inputs: a tanh perceptron $64 \to 96 \to 96$ trained on the sklearn digits dataset and frozen, with a linear head confined to a k-dimensional affine subspace $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{V}^\top \boldsymbol{c}$ under softmax cross-entropy, so the curvature is data-dependent and anisotropic and all dynamics occur in $\boldsymbol{c}$. The excitation is made r-dimensional rather than nominally so by modulating the loss weights of r of twelve fixed data groups with rationally independent sinusoids and orthogonalising the resulting gradient directions (Appendix E), after which the measured effective rank equals r to within 0.008 components. Six excitation regimes are run, spanning the three of table 1 at a fast and a slow drive period, with injected and with real mini-batch noise; two of them are results and not settings (Appendix E).

Recovery. Twelve observers were used, in five families: loss, norms, gradient, fixed random projections of the parameters, and function-space quantities. The estimator's five parameters were selected once and frozen, on four of those observers and on seeds and ranks reserved for the purpose, so what follows is out-of-sample in rank and seed throughout and in observer for six of the ten. On the withheld ranks and seeds the mean absolute error in the deterministic recurrent regime is 0.87 components run by run, and a median of 0.47 when the four seeds at each rank are averaged first, at a median slope of 0.92 and a perfect rank correlation for all ten. Errors by observer are in Appendix C; the parameter norms, which are what a run ordinarily logs, sit in the middle of the range and are also the two observers whose roughness is not flat against the rank, so part of their ordering may be spectral rather than geometric. The instantaneous mini-batch loss is excluded from every aggregate despite scoring among the best three, because it alone fails requirement 4: with the learning rate set to zero the parameters do not move at all, so there is no orbit and nothing to estimate, and eleven of the twelve observers are exactly constant, while its estimate still rises with the nominal rank from 0.91 to 6.93 because it contains the loss weights and therefore the drive.

Twenty directions. Repeating the design with twenty available directions gives the same picture with a lower ceiling: recovery is close to exact to about six directions and then saturates at 15.1 against a true 20.0, the ordering surviving at Spearman $+0.97$. That result is exploratory and we do not count it as validation, since its configuration was scored on the seeds it was selected on, the spread across seeds above eight directions is comparable to the error, and its Theiler exclusion is smaller

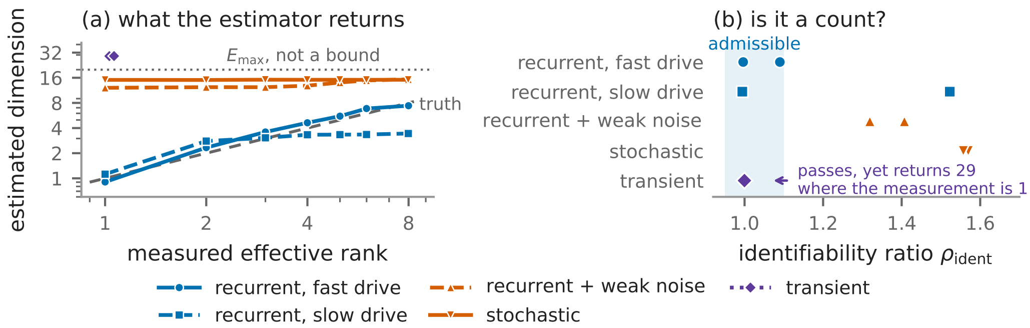

Figure 1: The evaluation of section 5.3: a tanh network trained on sklearn digits and frozen, a linear head confined to ten directions of which r are excited, five excitation regimes. (a) The estimate against the measured effective rank, which the construction holds at r, median over four seeds and ten observers; it tracks the truth only under a fast deterministic recurrent drive. (b) The identifiability ratio on the same runs, which separates those regimes without a ground truth. The transient is the one regime it passes, hence the recurrence count of section 3.4.

than its own embedding span (Appendix K). A second construction in function space (Appendix E) is the most accurate row of table 3, but its rank is imposed by an external teacher.

Table 3 collects the outcome; only the image-data rows carry a recovery error that may be read as one, the rest being listed so that the controls rejecting them are documented. The picture is otherwise consistent: recovery holds to about six components and degrades gradually above it, the ordering holds throughout, and the error on a real network with a real optimiser is of the same order as on a prescribed oscillation with none.

6 Conditions of validity

6.1 Dynamical regime

Figure 1 evaluates the estimator in each regime of table 1 at the same nominal rank, observer and matched octave. Under stochastic excitation it is flat in the rank, responding to $E_{\max}$, the correlation time and the record length, and the error exceeds ten components. Weak noise added to a recurrent trajectory destroys the absolute value while leaving the ordering: at a quarter of the pure noise amplitude, injected in the same r directions and leaving the measured effective rank at r, the error rises tenfold while the Spearman correlation stays above $+0.96$. A decaying transient, which returns near no state it has left and so has no active dimension at all, gives a large reproducible estimate above 27, and is the one regime ρ_{ident} does not catch. That the transient exceeds $E_{\max}$ is not the failure of a bound: no clamping is applied (section A), and a maximum likelihood fit returns values above the embedding dimension when its model is violated.

6.2 The delay lag

The delay window spans $(E_{\max} - 1)\tau$ samples and must cover a real fraction of the oscillation period, or the delay coordinates are collinear and the torus does not unfold; that the window rather than the lag is the quantity that matters is standard (Casdagli et al., 1991). On a fixed torus of period 400 at eight active directions, sweeping τ alone with the system unchanged moves the estimate from 2.60 to 63.35, with 8.53 at the span of four tenths of a period that is right (Appendix D). The correct lag is a function of the oscillation period, which a controlled experiment knows and a training log does not, and we do not solve this. The lag applied in section 7 is the frozen $\tau = 4$ of section B, transplanted from a calibration system of drive period 16, while the autocorrelation time of the parameter norm on those logs is 161 to 858: a matched lag would have been an order of magnitude larger. That reason for not reading the absolute value there would survive even if the dynamical regime were right.

6.3 Nuisance factors

At a fixed active dimension of four we vary seven factors that leave it unchanged, over five seeds and twelve observers, and compare the shift against a real four-component change (Appendix D). Two results matter. The smoothness control passes on eight of the ten observers, an octave shift of the drive band changing the roughness fourfold while moving the estimate only to the level of seed-to-seed variability; it fails on the two parameter norms, whose roughness orders the withheld ranks perfectly, so for the cheapest observer a run logs a smoothness explanation cannot be excluded. And a *ramped* observer gain, unlike a constant rescaling, moves the estimate by about a third of a real four-component change, so a run whose parameter norm grows drifts by order one component with nothing dimensional occurring: the nuisance that matters most in section 7.

6.4 The estimand is a count, not an effective rank

Every construction in section 5 makes the two quantities of section 3.1 numerically equal, by equalising the drive amplitudes so that the effective rank of an r-torus is r as well, and an experiment in which they agree cannot say which the estimator reports. Driving mode l at amplitude ρ^l separates them: the orbit still closes onto an r-torus, so $d_{\mathrm{act}} = r$, while the effective rank falls to $(\sum_l a_l^2)^2/\sum_l a_l^4$. The estimate follows the count: at four active directions the rank falls from 4.00 to 1.65 as ρ goes from 1 to 0.5 while the estimate stays between 3.74 and 4.06, and at two directions the rank falls to 1.47 while the estimate stays near 2.3.

It counts the components it can *resolve*: at eight directions, where the weakest mode is driven at under a hundredth of the strongest, the estimate falls with the effective rank from 8.88 to 4.86, the torus it can see having fewer dimensions than the one built. Over the grid it is closer to the count, 1.02 against 1.67 (Appendix D).

The diagnostics. Using no ground truth, the identifiability ratio separates the regimes in which the estimate is accurate, where it lies between 1.00 and 1.09, from those in which it is not, where it lies between 1.32 and 1.57; under the slow drive it separates within a regime too, reading 0.99 where the estimate is correct and 1.52 where it has saturated. It is not sufficient alone, reading 1.00 on the transient regime, and must be paired with the trend-crossing count, which is near zero there. Both are calibrated on these regimes only. The degeneracy flag fires on eleven of the thirty-five run-by-column cells of the logs of section 7.

7 Application to grokking

Grokking (Power et al., 2022) is studied in two experimental settings that differ in every relevant respect, and we work in both (Appendix L lists every run and every departure from the source configurations). The first is the one-layer transformer of Nanda et al. (2023) with AdamW and cross-entropy at weight decay 1.0, on modular addition and on S_5 composition (Power et al., 2022) at weight decay 0.2; we train it on mini-batches of 256 where the source is full-batch, which is what makes this the stochastic setting of table 1. The second is the unregularised full-batch gradient descent of Gromov (2023), a two-layer quadratic perceptron on mean squared error, run on the modular polynomials of Doshi et al. (2024); these supply label-matched target pairs differing by one monomial, one of which generalises while both fit the training set completely (Appendix J). Doshi et al. (2024) obtain their own results with Adam and heavy weight decay; we take the tasks and the representability criterion, not the optimiser.

7.1 The trajectory simplifies at generalisation

The effective rank of section 3.1 needs no delay reconstruction when the trajectory itself can be stored, and unlike d_{act} it is defined whether or not the orbit recurs, which here it does not. We store a sketch of the trajectory, in parameter and in function space, at every logged step of a transformer trained to grok, and report $\mathrm{PR}^{\mathrm{det}}$ (Appendix G). Near generalisation it collapses (figure 3): in function space by a factor between 2.9 and 15.6, median 6.6, reaching bottom 235 to 712 steps after the generalisation step, and in parameter space to between 1.09 and 1.60 one to eight hundred steps later again, both then re-expanding above the level they started from. The centred 600-step window is why “near” cannot be sharpened to “at”. Three observations support the reading: it sits at generalisation in seeds whose generalisation step differs by a factor of 3.7 and in a second task, so it is not a function of training progress; the displacement reaches its floor three thousand steps later and never recovers while the rank does, so it is not the trajectory stopping; and the two runs without weight decay fall by only 1.5 and 1.6.

That the solution passes through a simpler state at the transition is what the mechanistic accounts of grokking expect: Nanda et al. (2023) describe a cleanup phase in which the memorising components are removed once the generalising circuit has formed, and Varma et al. (2023) argue the transition happens because that circuit is the more efficient of the two. Both track what the weights encode as training proceeds; this is a statement about the geometry of the trajectory itself, and they agree. Appendix J qualifies it from the same runs: the Fourier order parameter of the representation is still rising $87\,000$ steps later, so the representation is not finished when the collapse occurs. None of this measures d_{act}, which is undefined for a transient; what it supports is that the variation of the trajectory within the window concentrates on one direction, not that the orbit lies on a one-dimensional set.

7.2 Without weight decay the function settles and the norm does not

The two weight-decay-zero controls never rise above chance validation accuracy, and what they do instead shows how little a low effective rank means alone. After memorisation both settle: the function-space effective rank falls from 28.6 to about 6 in modular addition and from 19.3 to 8.1 in S_5, and stays there. Their parameter norm meanwhile never comes back down, saturating at 1.4 times its initial value in the first and growing steadily to 2.4 times in the second, where in every generalising run it peaks and then falls by more than half as weight decay removes what memorisation built.

A low function-space rank is therefore reached by two quite different runs, one that has stopped moving and one about to generalise, and what separates them is the shape of the fall: the generalising runs hold a high rank through the memorisation plateau, collapse within a few hundred steps of the transition and come back up, while the controls decline once and stay down. Their growing norm is also the ramped gain of section 6.3, the mechanism behind the scalar-log fall of section 7.3.

7.3 What a scalar log adds, and on what scale

Applying the frozen configuration of section 5.3 to the logs of both settings gives table 11, which figure 2 plots. Two of its parameters cannot be carried across unchanged: the frozen window of $8\,000$ samples admits only three placements in a $12\,000$-sample record, so the window is reduced to a third of each record and the stride to $1\,000$, giving nine windows per transformer log and seven per perceptron log. Everything else, including $E_{\max}$, τ and the neighbour count, is the frozen configuration. The two settings occupy different regimes of table 1 and the diagnostics report which, without reference to any label. The regularised stochastic runs sit in a band of the identifiability ratio, 1.29 to 1.60, that the deterministic regimes never reach, with the roughness and trend-crossing count of a noisy series, and their estimates span a factor of four across seven runs: what figure 1 predicts for a stochastically excited series, set by $E_{\max}$ and the record and not by a rank. The unregularised full-batch runs sit at an identifiability ratio of unity with a roughness of one part in a thousand, two trend crossings per window and a delay participation ratio of exactly one, the signature of the transient regime. In neither is the value a count.

The series are not thereby useless. The level is uninterpretable because $E_{\max}$, the lag and the record length set it, and those are fixed within a run: one observer, one frozen configuration, one series. A *change* inside a run is therefore read against a fixed nuisance, which section 6.3 measures at about a third of a real four-component change for the largest of them. Two conclusions follow, pointing opposite ways: a change that is fast, localised and larger than that floor carries information the level does not, while a slow drift over a whole run does not, being what the growing norm of section 7.2 produces through the observer.

Label-matched pairs. The perceptron setting supplies controls the first cannot construct: both members of a pair reach full training accuracy under identical optimisation on identical data budgets and differ by one monomial in the target, so anything separating their logs is a property of the task–solution pair, not of the optimiser. On the training loss the estimate is about three units higher for the non-generalising member in all four pairs, with no overlap (Appendix F). It is not a dimension:

all eight runs are in the transient regime, and the mechanism is visible in the standard log, the representable targets driving the training error to zero while the perturbed ones stop three orders of magnitude short despite full training accuracy.

The direct measurement returns nothing here, for a reason that is a resolution limit and not a result: over a 600-step window a deterministic trajectory is straight to within the measurement precision. At 12 000 steps it has range, and a maximum of the same size appears in the generalising run and in its control (Appendix H).

What a fall is evidence of. A fall on a scalar log is not by itself specific to generalisation: one occurs in a 200 000-step run at $p = 211$ with no weight decay that never leaves chance accuracy, and it is slow, of the kind section 7.2 attributes to the norm. What the logs support is the weaker claim, that a fall on a log is not evidence of generalisation; the stronger one belongs to section 7.1 (Appendix I).

8 Limitations and conclusion

In a deterministic recurrent system with a matched delay lag, one scalar log of a suitable observer counts the independent components of the set the optimiser visits, to under one component up to eight of them and ordinally to twenty; outside that regime the set does not exist and what is returned is not a dimension. What makes the number usable is the diagnostics beside it, which flag those regimes without a ground truth. Applied to grokking they place both standard settings outside the admissible regime, while the trajectory effective rank, defined whether or not the orbit recurs, collapses by a median factor of six at generalisation and re-expands, in four regularised runs and not in their controls. Four limits bound this. The count needs a recurrent regime and a matched lag, and is ordinal above eight components. The collapse rests on four runs against two imperfectly matched controls, is located only to within half a window, is a signature and not an early warning, and does not reproduce in the second setting. And the diagnostics flag the regimes we could construct, not all of them.

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A THE ESTIMATOR

Algorithm 1 Pooled Levina–Bickel intrinsic dimension of a delay reconstruction

Require: series $x_{1:n}$; lag τ; embedding dimension E; neighbours m; Theiler exclusion W_T; dither σ

$z \leftarrow (x - \operatorname{mean} x)/\operatorname{std} x$; $\quad z \leftarrow z + \sigma\epsilon$, $\epsilon_t \sim \mathcal{N}(0,1)$ ▷ breaks exact ties
$\boldsymbol{y}_t \leftarrow (z_t, z_{t-\tau}, \dots, z_{t-(E-1)\tau})$ for $t = (E-1)\tau + 1, \dots, n$
$S \leftarrow 0$; $N \leftarrow 0$; $n_{\text{dist}} \leftarrow 0$; $n_{\text{ratio}} \leftarrow 0$
for each $\boldsymbol{y}_i$ **do**
  $r_1 \le \dots \le r_m \leftarrow$ distances to the m nearest $\boldsymbol{y}_j$ with $|i-j| > W_T$
  $r_j \leftarrow \max(r_j, \varepsilon_{\text{dist}})$; count those clipped into n_{dist}
  $S_i \leftarrow \max\big(\sum_{j=1}^{m-1} \log(r_m/r_j),\ \varepsilon_{\text{ratio}}\big)$; count clips into n_{ratio}
  $S \leftarrow S + S_i$; $\quad N \leftarrow N + 1$
end for
flag the window degenerate if n_{dist} or n_{ratio} exceeds $1\,\%$ of its denominator
return $\big(N(m-1) - 1\big)/S$

Three details of the implementation are not cosmetic. The returned value is $(N(m-1)-1)/S$ rather than $N(m-1)/S$, which is the exactly unbiased pooled maximum likelihood estimate for a $\Gamma(N(m-1), 1/d)$ sample and differs from equation (4) by a factor $1 - 1/(N(m-1))$, below 10^{-5} at the sizes used here. A flagged window is still returned rather than discarded; windows are dropped only when a series is summarised over its sliding windows, which is where every number in this paper is formed. And no clamping of the returned value to E is applied, so a divergent estimate is reported as divergent: clamping would convert the transient regime of table 1 into a plausible number.

The floors are $\varepsilon_{\text{dist}} = 10^{-8}$ on a neighbour distance and $\varepsilon_{\text{ratio}} = 10^{-5}$ on the per-point log-ratio sum, with $\sigma = 10^{-9}$ on the standardised series. The Theiler exclusion is $W_T = \max\big((E-1)\tau,\ t_{\text{acf}}\big)$, where t_{acf} is the first lag at which the autocorrelation falls below $1/e$, capped at 150 samples because the neighbour query cost is linear in it. Section K measures what that cap costs.

B FROZEN CONFIGURATIONS

Two estimator configurations are used in this paper, one for each range of the active dimension. Both were selected once, on data withheld from every later experiment, and then frozen. Table 2 states them and the grids they came from.

Table 2: The two frozen configurations and their selection. Only the swept parameters were free; the remainder were fixed before any estimate was computed.

	eight-direction	twenty-direction
$E_{\max}$	20	40
τ	4	16
m (neighbours)	20	20
Theiler window	autocorrelation	autocorrelation
window / stride	8 000 / 2 000	8 000 / 4 000
dither	10^{-9}	10^{-9}
swept	$E_{\max}$, τ, m, Theiler, window	$E_{\max}$, τ
grid size	48	4
selection seeds	90, 91, 92	0, 1, 2
selection ranks	2, 4, 6	2, 6, 10, 14, 18
withheld seeds	0–4	none
withheld ranks	1, 3, 5, 8	the other fifteen
selection error, best	0.324	2.47
selection error, worst	1.499	3.35

The selection objective in both cases is the error of the median estimate against the median measured effective rank, over the four observers reserved for selection, one from each observer family. The eight-direction objective adds a penalty on the rank correlation and on the spread across seeds; at the twenty-direction configuration both penalties are identically zero, so the objective there reduces to the error alone.

Four properties of these choices should be stated with them.

Both sit at a grid boundary. The eight-direction configuration is at the top of all five swept axes and the twenty-direction one at the top of both. That is less alarming than it sounds for the first, since four of its five axes have only two levels, so every configuration in that grid is at a boundary in four of them; τ is the only axis with an interior value, and the winner is at its top end. It does mean neither configuration is interior, and that a wider grid could move the level.

The selection error spans a factor of five on identical data. Across the eight-direction grid the error runs from 0.324 to 1.499 on the same logs, so the level is a choice as much as a measurement, and only the withheld numbers of section 5.3 may be read as recovery.

The Theiler axis is inert on this data. The autocorrelation rule returns the larger of the embedding span $(E_{\max} - 1)\tau$ and the measured autocorrelation time. On these logs the autocorrelation time is one to three samples while the embedding span is nine or more, so the two settings return bit-identical values at all 24 points of the grid where they are compared. The 48-point grid therefore contains 24 distinct measurements, and the choice of Theiler rule is not a choice at all here.

Neither configuration always uses the window its parameters name. At $E_{\max} = 40$ and $\tau = 16$ the requested exclusion is 624 samples, which the implementation truncates to its cap of 150 for the cost reason given in Appendix A. The eight-direction configuration requests 76 and is uncapped on every system of section 5, but on the training logs of section 7 the autocorrelation branch of the rule asks for 161 to 858 and is likewise truncated to 150. Appendix K measures what the truncation costs.

C Results by observer, and the alternatives

Table 3: The systems, ordered by increasing realism. The last column names the requirements of section 4 each row fails; only the image-data parameter-subspace row at 1–8 meets all six, and it is the row the recovery claim of this paper rests on. The unit of analysis is one (rank, seed, observer) run, each side of the error being the median over that run's sliding windows, except where noted; ρ is over ranks after a median over seeds. The two oscillating-matrix figures are its two held-out seeds, that system having one observer.

system	truth	range	MAE	ρ	protocol
oscillating matrix	constructed	1–10	0.30 / 0.69	0.99–1.00	fails 2, 5
online linear regression	constructed	1–20	1.34	0.980	fails 2
logistic regression	constructed	1–20	1.33	0.986	fails 2
frozen nonlinear decoder	constructed	1–20	2.13	0.924	fails 4
perceptron in a k-subspace	constructed	1–20	2.06	0.939	fails 4
image data, parameter subspace	measured	1–8	0.87	$+1.000$	**complete**
image data, parameter subspace	measured	1–20	1.62	$+0.973$	fails 3
image data, function subspace	measured	1–8	0.50	$+0.988$	fails 2, 3, 4

Two rows of table 3 predate the estimator of Appendix A and are marked in that table with the requirements they fail. The oscillating matrix was scored with an earlier implementation that applies no Theiler exclusion and clamps the returned value to E, at $E = 25$ and $m = 30$ rather than at the frozen configuration, and its grid swept the drive period alongside the estimator parameters, which is the breach of requirement 2 that also disqualifies the two regression systems; it reports no companion statistics, which is requirement 5. The function-subspace row shares the clamping and the mixed grid, computes one window per run rather than a median over sliding windows, and has no zero-learning-rate control, so requirement 4 is untested there rather than passed. Both rows are kept because they locate the ceiling and because the oscillating matrix supplies the synchronisation control of section 5.1, which no other system provides; neither should be read as a recovery result.

All numbers in this appendix are computed on the withheld ranks $r \in \{1, 3, 5, 8\}$ and the withheld seeds of section 5.3, in the deterministic recurrent regime, and are scored against the measured effective rank rather than against r. Where an aggregation over seeds is needed, the median over the four seeds at each rank is taken before scoring; that is the convention the analysis uses throughout, and it is stated because the choice moves the numbers. Table 4 gives the result for every observer.

Table 4: Per-observer results on withheld ranks and seeds. The last column is the rank correlation between the roughness statistic and the measured effective rank: a value near zero means the ordering that observer produces is not available from smoothness alone.

observer	family	MAE	slope	ρ	ρ(roughness)
probe loss	loss	0.303	0.904	$+1.000$	$+0.200$
function-space norm	norm	0.361	0.931	$+1.000$	$+0.200$
margin	function	0.409	0.931	$+1.000$	$+0.200$
parameter norms (two)	norm	0.448	0.849	$+1.000$	$+1.000$
fixed parameter projection	projection	0.489	0.966	$+1.000$	$+0.200$
gradient projection	gradient	0.559	0.891	$+1.000$	$+0.800$
function-space projection	function	0.699	0.705	$+1.000$	-0.200
gradient norm	gradient	1.972	1.654	$+1.000$	-0.800
full-batch loss	loss	2.008	1.621	$+1.000$	-0.800
instantaneous loss	loss	0.418	0.863	$+1.000$	0.000
probe accuracy	function	degenerate in every window			

Two observers are excluded from every aggregate, for different reasons. Probe accuracy is quantised, which drives the delay vectors together and trips the degeneracy flag in $100\,\%$ of its windows; it is the sole source of the degenerate fraction reported anywhere in this study, every other observer being at zero. The instantaneous mini-batch loss is excluded because it fails requirement 4: at zero learning rate, where the parameters do not move at all, its estimate still reads 0.91, 2.22, 3.71, 4.62, 5.45, 5.31 and 6.93 across $r = 1, \ldots, 8$, a range of six components produced entirely by the drive. The other eleven observers are exactly constant under that control, so no estimate is defined for them.

The roughness column carries a caveat that the main text states and that this table makes precise. For seven of the nine usable observers the roughness statistic carries little or no ordering, so their agreement with the truth is not available from smoothness. For the two parameter norms it orders the withheld ranks perfectly, and those are exactly the observers a training run ordinarily logs. Their result should therefore be read as consistent with, rather than as evidence against, a smoothness explanation.

Table 5: Twenty available directions, image data, median over three seeds and eight observers. The truth is the measured effective rank; the last row is the spread of MG across seeds at fixed rank.

true r	1	2	4	6	8	10	12	14	17	20
MG	0.89	2.35	4.41	6.66	7.38	8.99	10.61	11.40	13.50	15.09
$\mathrm{PR}_{\mathrm{delay}}$	2.00	2.29	4.54	6.81	7.95	9.20	10.79	12.15	11.71	13.51
spectral PR	1.01	1.25	2.58	3.68	4.18	5.48	5.98	6.53	8.11	8.38
spread across seeds	0.00	0.20	1.49	2.15	2.11	2.95	1.50	2.84	2.76	3.33

Table 6: The estimator against the alternatives, on the same data and under the same convention, at both ranges. Statistics marked † need a neighbour search; the rest are one-line computations on the series or its spectrum.

	eight directions		twenty directions	
statistic	MAE	ρ	MAE	ρ
MG †	0.457	$+1.000$	1.632	$+0.961$
LB †	0.435	$+1.000$	1.358	$+0.979$
TwoNN †	1.814	$+1.000$	3.296	$+0.864$
delay participation ratio	0.894	$+1.000$	2.320	$+0.982$
spectral PR, 256 bands	1.604	$+1.000$	5.401	$+0.996$
spectral PR, 1024 bands	1.353	$+1.000$	4.547	$+0.996$
spectral PR, native resolution	1.026	$+0.800$	2.760	$+0.968$
roughness	3.652	$+0.200$	10.069	-0.482

Table 6 should be read with three things in mind, and none of them favours MG.

First, MG is not the most accurate statistic at either range. The uncorrected per-point average is ahead of it at both, by 0.02 components at eight directions and by 0.27 at twenty. The correction of section 3.2 removes a positive bias, and on a saturating estimate that bias was partly cancelling the saturation, so removing it costs accuracy exactly where the estimate is already saturating.

Second, MG is not the best-ordering statistic at twenty directions. The spectral participation ratio at 256 and 1024 bands orders the ranks more faithfully than MG does, at four times the error. If only the ordering is wanted, a statistic requiring no embedding and no neighbour search will supply it.

Third, MG's advantage over the linear delay participation ratio is a factor of about two at eight directions, not the factor of six that the selection split of Appendix B reports. That larger figure is computed on the ranks and seeds the configuration was chosen on and is a property of the selection, not a recovery result.

The advantage depends on the observer. Running the same torus through two scalar observers that differ only in how evenly they weight the modes settles which statistic is preferable, and the answer is that it depends. Under an observer that weights all modes equally, the spectral participation ratio at 256 bands attains an error of 0.35 against MG's 1.19, ahead by 3.4 times. Under a generic observer with random weights, which is what a random projection of a network's parameters actually is, MG attains 0.34 against 1.61, ahead by 4.7 times. Both orderings are perfect in three of those four cells, so what separates the statistics is absolute recovery and not rank agreement. Both figures also use the delay lag that is best for that observer, which a controlled experiment can choose and a training log cannot: at a lag common to both observers the two factors become 6.6 and 1.9 respectively.

D Conditions of validity in detail

The delay lag. The delay window spans $(E_{\max} - 1)\tau$ samples, and table 7 sweeps τ on a fixed torus of period 400 at $E_{\max} = 20$, leaving the system unchanged. At a span of one twentieth of a period every rank returns the same tangent constant near 2.5; at four tenths the estimate follows the rank to within one component; at one and a half periods it exceeds the rank eightfold. The estimate at $r = 8$ moves by 61.5 components across the sweep and the estimate at $r = 1$ by 0.03, so the failure is not a uniform bias but a collapse of the ordering at one end and a divergence at the other.

Table 7: Estimate against the delay lag on a fixed torus of period 400, $E_{\max} = 20$. The span is the delay window in units of the oscillation period. The correct rank is the column heading.

τ	span	$r = 1$	2	3	4	6	8
1	0.05	1.42	2.44	2.49	2.63	2.44	2.60
4	0.19	1.43	2.72	3.31	3.83	3.94	4.29
8	0.38	1.43	2.77	4.18	6.16	6.57	8.53
16	0.76	1.44	3.00	5.92	13.66	12.83	20.76
32	1.52	1.46	3.06	9.90	33.28	31.02	63.35

Nuisance factors. Table 8 varies seven factors at a fixed active dimension of four, over five seeds and twelve observers, against the real four-component change measured below. Rotating the coordinates is scored on the three projection observers alone, since the remaining nine are invariant to it by construction, so it measures a property of the observer rather than of the estimator.

Table 8: Shift in the estimate produced by factors that leave the active dimension unchanged. The first row is the seed-to-seed variability of an unchanged configuration and sets the scale against which the others should be read.

factor	$\lvert\Delta\mathrm{MG}\rvert$	as % of a real four-component change
none (baseline)	0.21	5
learning rate halved	0.16	4
drive band up one octave	0.49	12
drive band down one octave	0.50	13
coordinates rotated	0.86	22
observer gain ramped $10\times$	1.16	29
state amplitude ramped $4\times$	1.32	33

Detecting a change. Recovery of a static level does not establish that a change is detected. Running three consecutive segments $r_{\text{high}} \to r_{\text{low}} \to r_{\text{high}}$ with nothing else altered, and re-measuring the ground truth on each segment, a true fall of 3.99 components is observed as 3.95 in the deterministic recurrent regime, is detected by every usable observer in both directions, and recovers to the level before it to within 0.01 components. Under stochastic excitation the same true change is observed as 0.15.

The rule fires when the estimate falls below the median of the pre-switch segment by three times that segment's window-to-window scatter, for two consecutive windows. The threshold uses the pre-switch segment alone, never the midpoint of the two observed levels, which would hand the detector the answer and would decide the arm where the two levels coincide by a coin flip. On the parameter norm the level pairs read $4.77 \to 1.22 \to 4.78$ for a true $4 \to 1 \to 4$ and $6.90 \to 2.12 \to 6.91$ for a true $6 \to 2 \to 6$; at $8 \to 3 \to 8$ they read $6.97 \to 3.64 \to 6.97$, under-reporting the high level by a component, which is the saturation of table 5 seen from another direction. The level error is 0.89 at the high level and 0.22 at the low one: the estimate is more accurate the fewer directions are active.

The response time cannot be quoted from this experiment. With a window of $8\,000$ samples and a stride of $2\,000$ the two observed lags, 999 and $1\,999$ steps, are the two floors of the measurement grid rather than measured values. What the experiment does establish is that the estimate crosses the threshold on a window still containing $87.5\,\%$ pre-switch data.

Anisotropy. Table 9 gives the measurement of section 6.4 in full: the system of section 5.1 with mode l driven at amplitude ρ^l, three seeds, observed through the squared Frobenius norm and estimated with the frozen eight-direction configuration. The manifold dimension is r in every row. The measured participation ratio agrees with the closed form $(\sum_l a_l^2)^2/\sum_l a_l^4$ to two decimals throughout, which is the check that the construction does what it claims.

Table 9: An r-torus with geometrically decaying amplitudes. r is the manifold dimension and is constant along each block; PR is the participation ratio of the covariance of the r oscillating coordinates. Median over three seeds.

r	ρ	PR	MG	$\mathrm{PR}_{\mathrm{delay}}$
2	1.0	2.00	2.30	3.12
2	0.7	1.79	2.33	3.28
2	0.5	1.47	2.35	3.18
4	1.0	4.00	3.82	6.44
4	0.7	2.60	3.89	4.85
4	0.5	1.65	4.06	3.11
6	1.0	6.00	8.76	8.23
6	0.7	2.84	6.90	5.53
6	0.5	1.67	4.83	3.21
8	1.0	8.00	8.88	8.76
8	0.7	2.90	5.26	5.19
8	0.5	1.67	4.86	3.00

At two and four directions the estimate is flat in ρ while the participation ratio falls by a factor of 1.4 and 2.4, which is the clean case. At six and eight it falls too, and the reason is visible in the amplitudes: at $\rho = 0.5$ and $r = 8$ the weakest mode is driven at 0.008 of the strongest, far below the neighbourhood scale the estimator works at, so the torus it sees really does have fewer dimensions than the one that was built. The estimator is therefore counting modes it can resolve, which is neither the manifold dimension of the construction nor the participation ratio of its covariance, and coincides with both only when the amplitudes are equal.

E System constructions

The image-data system of section 5.3. The data is partitioned into twelve fixed groups, the same partition at every rank so that the partition is not itself a function of r, and the loss weights of r of them are modulated by r rationally independent sinusoids filling one octave. Modulating group j tilts the gradient along a direction ϕ_j. These directions are neither orthogonal nor of equal effect, since random data groups have correlated gradients, so a mixing matrix $\mathrm{pinv}(\Phi)Q$ with Q an orthonormal basis of $\mathrm{range}(\Phi)$ orthogonalises them, and the per-mode scalar response gain of the linearised update is divided out. Without this correction the trajectory participation ratio falls well below r.

Table 10 verifies that the construction achieves what it intends, and two of its rows are results rather than settings. A decaying transient has no recurrent set whatever r is, since a converging trajectory is a curve; and plain mini-batch descent sits at 6.2 regardless of r, because its noise rank is fixed by the per-example gradient covariance of the data and is not a free parameter of the experiment. Projecting real mini-batch noise onto r directions recovers only part of the intended rank, reaching 5.6 at $r = 8$, which is why that regime is scored against its measured value and not against r.

Table 10: Measured trajectory participation ratio by excitation regime and nominal rank, median over seeds, for the ten available directions of section 5.3. Every estimate in this paper is scored against these values, never against r.

regime	$r = 1$	2	4	6	8	what is excited
`qp`	1.00	2.00	4.00	6.00	8.00	r sinusoids, fast drive
`qp_slow`	1.00	2.00	4.00	6.00	7.99	the same at a training-log timescale
`qp_nopre`	1.00	1.99	3.99	5.89	7.55	`qp` without the preconditioner
`mixed`	1.00	2.00	4.00	6.00	7.99	`qp` plus noise in the same directions
`noise`	1.00	2.00	3.99	5.99	7.98	injected rank-r gradient noise
`noise_nopre`	1.10	2.08	3.74	5.47	6.43	the same without the preconditioner
`batch_proj`	1.00	1.79	3.25	4.53	5.61	real mini-batch noise projected to rank r
`batch`	6.20	6.25	6.21	6.26	6.20	plain mini-batch descent
`gd`	1.05	1.05	1.05	1.03	1.06	a decaying transient
`qp`, $\eta = 0$	1.00	1.00	1.00	1.00	1.00	requirement 4 control

The last row is the control of requirement 4. With the learning rate multiplied by zero the active dimension is identically one at every nominal rank, so an observer whose estimate still tracks r there is reading the drive rather than the optimiser.

The functional rank is 10 at every rank, seed and regime, with a participation ratio between 8.40 and 9.03 across the four seeds; it is a property of the frozen backbone and does not depend on the excitation. The condition number of the drive-direction matrix is 1.00 at $r = 1$ by construction and has a median of 13.3 at $r = 8$, which is the anisotropy the mixing matrix exists to remove.

The function-space construction. A local adapter is built around a trained $64 \to 16 \to 10$ network and its function-space Jacobian is QR-whitened, so that its first r columns have rank r at equal local scale. A quasiperiodic teacher excites every adapter coordinate and the student tracks it by exact backpropagation through both layers. The functional, trajectory-covariance and update-covariance ranks were all verified to equal r.

The twenty-direction system. The same construction at twenty available directions reproduces the nominal rank to within 0.008 components over $r = 1, \ldots, 20$ in the recurrent regime, and to three decimals up to $r = 10$. The other regimes behave as at ten directions: injected noise tracks r closely, projected mini-batch noise reaches only 9.4 at $r = 20$, and the decaying transient stays between 1.02 and 1.09 throughout.

F Figures for the grokking application

Table 11: The estimator and its diagnostics on training logs at $E_{\max} = 20$: the regularised stochastic setting through the parameter norm, the unregularised full-batch setting through the training loss. Runs that have not generalised within the budget are censored, not negative.

run	generalises	MG	ρ_{ident}	roughness	PR_{delay}	crossings
transformer, positive control	yes	15.31	1.289	0.030	n/a	56
transformer, reduced data, s0	yes	4.93	1.599	0.068	n/a	146
transformer, reduced data, s2	no	4.47	1.514	0.149	n/a	151
transformer, no weight decay	no	28.08	0.999	0.000	n/a	2
perceptron, $n + m$	yes	19.54	0.999	0.001	1.0	n/a
perceptron, $n \cdot m$	yes	19.63	0.999	0.001	1.0	n/a
perceptron, $n^3 + nm^2 + m$	no	22.43	0.998	0.000	1.0	n/a

Figure 2 places every training log of section 7 in the plane of the two diagnostics, against the region in which every accurate case of section 6 fell. No run of either setting is in it: the regularised mini-batch runs are unstable in the embedding dimension, and the full-batch and zero-weight-decay runs are stable only because their series is monotone.

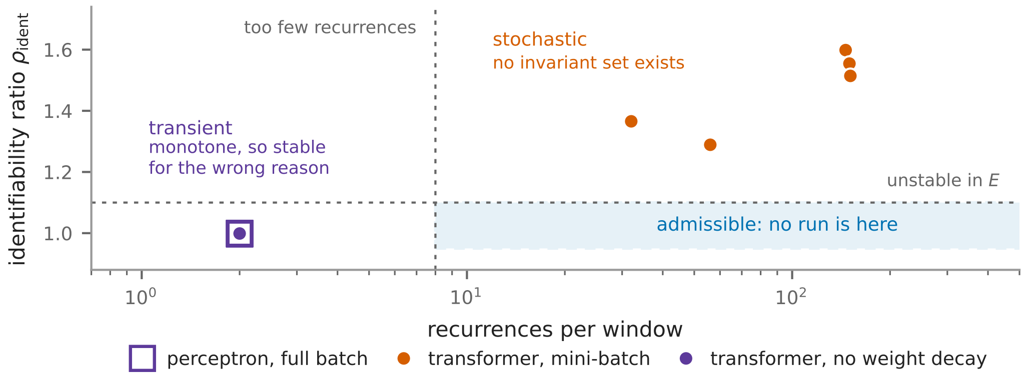

Figure 2: Every training log of this section in the plane of the two diagnostics of section 3.4, one point per run: seven transformer runs of 120 000 steps observed through the parameter norm, and ten perceptron runs of 100 000 steps observed through the training loss. The vertical axis is the identifiability ratio, near unity when the estimate does not depend on the embedding dimension; the horizontal axis counts how often the series crosses its own least-squares line inside a window. The shaded rectangle is where every accurate case of section 6 fell: $\rho_{\text{ident}} \leq 1.10$, which is the top of the range observed on the recurrent arms, and more than eight crossings, which is the bottom of the range on any arm the estimate tracked. It is a description of the calibration, not a validated decision rule, and no run of either setting is inside it. The two settings fail in different corners, and both failures are visible without any label.

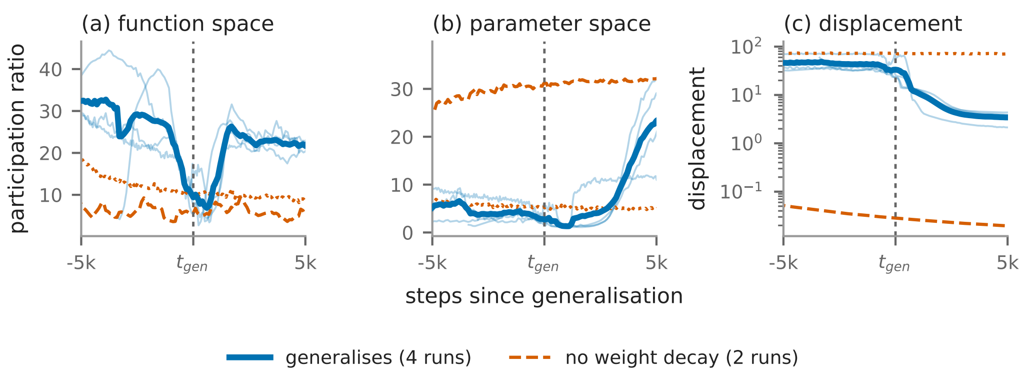

Figure 3: The trajectory effective rank measured directly (section 7.1), on six transformer runs: three seeds of modular addition at weight decay 1.0 and one S_5 composition at 0.2, against the same two configurations at weight decay zero, which memorise and stay at chance validation accuracy for their whole budget. The statistic is $\mathrm{PR}^{\det}$ of section 3.1, an effective rank; these runs have no recurrent set, so no active dimension is defined for them. Windows of 600 steps at their midpoint, so the window is not causal and a feature is located to within ± 300 steps; each control, having no generalisation step, is aligned on that of the matched weight-decay run (Appendix G). Thick blue is the mean of the four generalising runs, thin blue the runs themselves.

Figure 4 shows the separation of section 7.3 and the mechanism behind it. The estimate is about three units higher for the member of each pair that never generalises, in all four pairs and with no overlap, but the training loss shows that the two members differ in the shape of their curves long before any question of dimension arises.

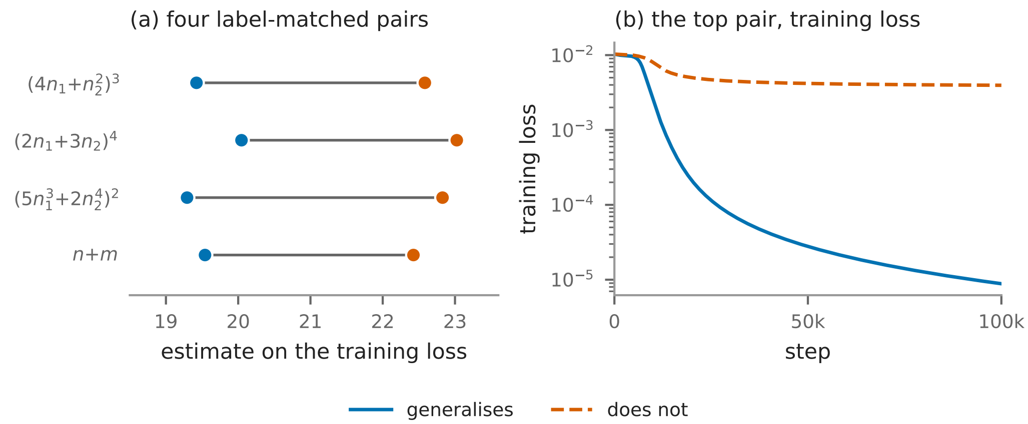

Figure 4: (a) The estimate on the training loss for the four label-matched pairs of section 7.3: three modular polynomials at $p = 97$, each against itself plus one monomial, and modular addition against $n^3 + nm^2 + m$. Both members of a pair reach full training accuracy under identical optimisation on identical data budgets. (b) The top pair of (a), and what the estimate is reading in it. The representable target drives the training loss to zero, while the perturbed one stops three orders of magnitude short despite fitting the training set completely, so the two curves differ in shape rather than in any attractor dimension.

G The trajectory sketch

The direct measurement of section 7.1 needs the trajectory itself, which for the modular transformer is 226 816 doubles at each of two to three thousand logged steps (228 608 for S_5). Every logged step is therefore compressed by a CountSketch (Charikar et al., 2002; Weinberger et al., 2009) to $\mathbb{R}^{1024}$, with two independent hash families. Two spaces are recorded: the parameters, and the probe logits after centring each example over classes and normalising to unit length. The normalisation is not cosmetic. The raw logit scale is mechanically coupled to weight decay, so a signal read off it could not be distinguished from a detector of weight decay.

We do not verify a spectral guarantee for this. The elementary bound for a sketch of this kind is on pairwise inner products (Weinberger et al., 2009), and a participation ratio is a functional of the whole spectrum, on which an entrywise Gram bound is weak. A subspace-embedding guarantee does exist for CountSketch (Clarkson & Woodruff, 2013) and would apply here, since a sixty-sample window has rank at most sixty, but it requires a rank we do not measure per window. What the two hash families supply instead is a direct estimate of the error: every reported value is the mean of the two, and their disagreement is recorded with it. Over the published transformer windows that disagreement is $1.1\,\%$ of the value at the median and $8.4\,\%$ at its worst, which is the number the reader should use, and it is smaller than every effect section 7.1 reads.

Two further checks stand behind the numbers. Table 12 pushes synthetic trajectories of known rank through the sketch and compares against the same statistic computed on the uncompressed trajectory. The sketch is accurate to 0.04 components throughout; the shortfall against the nominal rank at 5 and 10 is a property of the participation ratio on a sixty-sample cloud and is present without any compression, which is what separates the two questions. Nothing above rank ten is tested, and the plateaus in figure 3 are higher than that. On a real run of the perceptron architecture, where the parameter count is small enough to allow it, the participation ratio computed exactly on all $145\,500$ raw parameters agrees with the sketched value to three decimals. And the observer is verified not to perturb training by requiring the log to be bit-identical with and without it, over four hundred steps at reduced probe size, which matters because the task module seeds one global random stream that the train–validation split, the initialisation and the mini-batch order all continue: a single stray draw changes the initial weights and can destroy grokking.

Table 12: The sketch against the same statistic computed without it, on synthetic trajectories of 60 samples and known rank in $145\,500$ dimensions. The middle column is what the compression costs; the gap between the last two columns is what a sixty-sample participation ratio costs, and is not attributable to the sketch.

true rank	PR, uncompressed	PR, sketched	sketch − uncompressed
1	1.00	1.00	0.00
2	1.99	1.98	−0.01
5	4.42	4.46	0.04
10	8.61	8.63	0.02

Windows are sixty logged rows, which is 600 optimiser steps for the modular runs and 300 for S_5, and are labelled by their centre, so a feature at a given step is located to within half a window and the window is not causal. Five statistics are computed per window in each space, of which the paper reports two: $\mathrm{PR}^{\mathrm{det}}$, the participation ratio of the positions after removing a per-window linear trend, since a steady drift is rank one and would otherwise mask everything else; and $\mathrm{PR}^{\mathrm{upd}}$ on the increments block-averaged over five logged steps, to suppress the mini-batch noise floor. Beside them the total displacement is recorded, as the null for a rank that falls simply because the trajectory stopped moving.

H The full-batch setting

Repeating the direct measurement on the perceptron setting gives a result that must be read as a limit of the measurement rather than as a negative. Under full-batch gradient descent every window of every run has a participation ratio between 1.000 and 1.362 on the four statistics the comparison uses, and between 1.000 and 2.171 over all eight recorded columns. This is not an artefact of the sketch: computed exactly on all $145\,500$ raw parameters the value is the same to three decimals, and the sketch recovers synthetic trajectories of known rank up to ten. A deterministic update makes the trajectory a smooth curve, and over a 600-step window a smooth curve is straight to within the measurement precision, so what the statistic reports is the curvature of the trajectory at the scale the window sets, and at that scale there is almost none. It has no range in which a collapse could appear, and the absence of a feature is therefore not evidence of its absence: the window must be set in units of the curvature scale of the trajectory. Repeating the measurement against window length confirms the diagnosis and withdraws the opening it appeared to leave. Two runs of $150\,000$ steps, a generalising one and its label-matched control, are measured over windows spanning 600 to $120\,000$ optimiser steps, with the window lengthened by spreading sixty samples further apart rather than by adding samples, so that the bound on the statistic and its noise floor stay exactly where the published measurement put them and only the span changes. Table 13 gives the result. The statistic does leave the floor, at a scale of about 10^4 steps, two orders of magnitude above the window that produced the 1.000–1.362: the absence of a feature there was a resolution limit and not a fact about the run.

It is not, however, a hidden signal. What appears at $12\,000$ steps is one localised maximum, and both runs have one, of the same size: 1.877 centred at step $12\,800$ for the run that memorises at $9\,200$ and generalises at $11\,950$, and 1.843 centred at $9\,350$ for the control, which memorises at $9\,800$ and does not generalise within the budget. Function space agrees with more range and no more separation, 2.618 against 2.145. What cannot be concluded from this is where the maximum sits relative to either milestone. The two centres are adjacent points of a $3\,450$-step grid and their windows overlap over $6\,900$ to $15\,250$ steps, so a $12\,000$-step window cannot localise a feature to better than the interval between the two milestones; measured against memorisation the offsets are $+3\,600$ and -450, which is a difference the design cannot resolve. The claim the experiment does support is the negative one: at the window where the statistic has range, the generalising run and its label-matched control are not separated by it. Beyond $30\,000$ steps the ordering reverses and the control moves further than the generalising run, because the window is then a large fraction of a run whose weight norm is still climbing and the statistic is reading that drift. The full-batch trajectory therefore carries no rank signature of generalisation at any window length between 600 and $120\,000$ steps, which is the stronger form of the same conclusion: the collapse of section 7.1 belongs to the regularised stochastic setting, and a trajectory participation ratio must be controlled for window length as well as for batch size before anything is read from it.

Table 13: Detrended position participation ratio in parameter space against window length, full-batch gradient descent at $p = 97$, over all placements of the window. Sixty samples per window at every length. The 600-step row is the measurement reported above, on the same configuration and seed.

	generalises		control, never generalises	
window (steps)	median	max	median	max
600	1.002	1.035	1.001	1.011
3 000	1.070	1.253	1.003	1.092
6 000	1.028	1.537	1.005	1.360
12 000	1.011	1.877	1.020	1.843
30 000	1.007	2.111	1.113	1.725
60 000	1.021	1.660	1.335	1.975
120 000	1.256	1.659	1.929	2.337

Introducing mini-batches confirms the diagnosis. The same statistic then ranges over more than a decade while the generalisation step moves by one logged step, so the learning is unchanged and the statistic has moved by an order of magnitude. Any claim read from a trajectory participation ratio must therefore be controlled for batch size and noise scale before it is attributed to learning.

I What falls, and when

A fall of the estimate is not specific to generalisation. In a 200 000-step run at $p = 211$ with no weight decay, which memorises at step 2 250, ends at $3.6\,\%$ validation accuracy and grows its parameter norm from 43 to 126, a fall is detected at step 49 250, and the amplitude control of section 6.3 supplies a mechanism that needs no dimension at all: a ramped observer gain moves the estimate by about a third of a real four-component change.

What that run cannot show is that its trajectory simplified. Its log sits in the transient regime of table 11, where the estimate is not a dimension, so a fall in it is a fall of a quantity we have argued should not be read. The two must be kept apart: the direct measurement of section 7.1 is a dimension and falls in four generalising runs and not in their controls, whereas the estimate on a log, in these settings, is not a dimension and falls in at least one run that does not generalise. What follows is the weaker and defensible claim, that a fall on a log is not evidence of generalisation. Nor is the absence of generalisation within a budget evidence that a fall was a false alarm: two reduced-data configurations against which earlier specificity figures were computed generalise once the budget is extended, at steps 39 600 and 110 940, so such runs are censored and no specificity figure may be computed against them.

J The representation dimension in closed form

The perceptron setting of section 7 makes the learned representation analytically available, and three different numbers can be read off it. They are not commensurable and the literature does not always separate them. The *mode count* is $(p + 1)/2$, or 49 at $p = 97$: the number of distinct Fourier frequencies the generalising solution of Gromov (2023) uses. The *order parameter* is the mean inverse participation ratio of the first layer's Fourier spectrum, which is 1.000 for that solution because each neuron carries a single frequency and about 0.041 at random initialisation; it measures how concentrated each neuron is, not how many modes exist. The *effective rank* is the participation ratio of the singular values of the first-layer weight matrix, which for the analytic solution is 148.8: that matrix is 500×194 and full rank, because each non-zero frequency contributes a sine and a cosine direction in each of the two input blocks, so 148.8 says how evenly those 194 directions are used and is not a count of anything. Only the first is a dimension in the sense the closed form fixes. Two things follow from measuring the other two, and both qualify readings of the representation dimension elsewhere.

The representation is not complete at the transition. For modular addition the order parameter reads 0.067 just after memorisation at step 9 170, 0.189 just before generalisation at step 11 950, 0.263 when validation accuracy first reaches $100\,\%$, and then rises for a further 87 000 steps to 0.595 against an analytic 1.000. It does not jump at the transition; it begins rising at memorisation and passes smoothly through. Generalisation therefore arrives well before the representation is complete, and requires it only to be periodic enough. The effective rank of the same layer carries almost no signal at all: it reads 139.1 at random initialisation, within seven per cent of the analytic 148.8 before any training, then spans 132.6 to 150.9 over the run without being monotone and overshoots the reference before falling back. Landing near the reference is therefore not evidence of anything.

The order parameter is only interpretable against its own reference. Table 14 gives the measured value beside each task's own closed form. Two of the tasks that generalise have a reference near the floor, because the representation is periodic in $g_1(n)$ while the spectrum is taken over the raw index n, and g_1 is not linear there. Read against modular addition's 1.000, a measured 0.044 says nothing was learned; read against that task's own 0.062, it says the run has essentially converged. The tasks that never generalise sit at the initialisation floor, and the analysis returns no reference for them rather than substituting another task's.

Table 14: Fourier inverse participation ratio of the first layer at the end of training, against each task's own closed form. All runs are full-batch gradient descent with no weight decay at $p = 97$. A dash marks a task for which the source papers give no closed form.

task	generalises	measured	own reference	grokking step
$n + m$	yes	0.595	1.000	11 950
$n - m$	yes	0.582	1.000	11 570
$n^2 + m^2$	yes	0.093	0.052	10 380
$n \cdot m$	yes	0.073	—	11 600
$(4n_1 + n_2^2)^3$	yes	0.243	1.000	12 560
$(2n_1 + 3n_2)^4$	yes	0.283	1.000	7 750
$(5n_1^3 + 2n_2^4)^2$	yes	0.044	0.062	6 430
$(4n_1 + n_2^2)^3 + n_1 n_2$	no	0.040	—	none
$(2n_1 + 3n_2)^4 - n_1^2$	no	0.043	—	none
$(5n_1^3 + 2n_2^4)^2 - n_2$	no	0.039	—	none
$n^3 + nm^2 + m$	no	0.040	—	none

K The Theiler exclusion of the twenty-direction configuration

The Theiler exclusion is meant to remove neighbours that are close in time rather than in phase space, and the rule this study asks for is $W_T = \max((E-1)\tau,\, t_{\text{acf}})$: at least the full span of a delay vector, so that two neighbours cannot be near each other merely because their windows overlap. The implementation caps W_T at 150 samples, because the neighbour query cost is linear in it. At the eight-direction configuration the embedding span is $(20 - 1) \times 4 = 76$, so the cap binds only where the autocorrelation time exceeds 150: never on the calibration systems, whose autocorrelation time is one to three samples, but on five of the seven parameter-norm records of section 7, where it is 161 to 858. Those five were therefore scored at $W_T = 150$ rather than at the value the rule asks for, in the direction that inflates the estimate. At the twenty-direction configuration the span is $(40 - 1) \times 16 = 624$, so the cap binds unconditionally and the exclusion used is 150, under a quarter of what the rule asks for. Every row of the twenty-direction scores records $W_T = 150$; this is a departure from the stated protocol, not a choice we made deliberately, and it is why section 5.3 marks that result exploratory.

Table 15 measures what it costs, by re-scoring the stored trajectories at both exclusions. Nothing is regenerated, so the systems cannot drift between the arms.

Table 15: The twenty-direction system re-scored at the capped exclusion and at the one the configuration asks for. Seven ranks, three seeds, the three observers with the lowest error, median over seeds; the error is against the measured $\mathrm{PR}^{\mathrm{pos}}$ and is not comparable with table 3, which is taken over more observers. What matters is the difference between the columns.

r	measured $\mathrm{PR}^{\mathrm{pos}}$	MG at $W_T = 150$	MG at $W_T = 624$
2	2.00	2.35	2.24
5	5.00	4.53	4.11
8	8.00	5.82	5.34
11	11.00	9.16	8.67
14	14.00	9.56	9.04
17	17.00	13.16	12.71
20	19.99	13.64	13.45
median absolute error		2.78	3.13
Spearman ρ		$+1.000$	$+1.000$

Three things follow. The ordering, which is what section 5.3 claims at this range, is unaffected: the rank correlation is $+1.000$ under both exclusions. The absolute value moves down by 0.1 to 0.5 components, so the capped exclusion was mildly inflating it, in the direction that a shortage of exclusion predicts. And the correction makes the recovery slightly worse rather than better, 2.78 to 3.13 in median absolute error, so the cap was not flattering the result: enforcing the protocol costs accuracy and buys correctness. The saturation above about six directions is present at both settings and is therefore not an artefact of the exclusion.

The same cap has a second effect that the identifiability ratio inherits. The exclusion actually applied is $W_T = \min\bigl(\max((E-1)\tau,\ t_{\mathrm{acf}}),\ 150\bigr)$. The numerator is computed at $2E_{\max}$ with τ held fixed, so at the eight-direction configuration its span is $(40-1)\times 4 = 156$ and the denominator's is 76: the numerator is capped at 150 whatever the data does, while the denominator is capped only once t_{acf} reaches 150. The two arms are therefore treated identically exactly when $t_{\mathrm{acf}} \geq 150$ and differently when it is smaller. On the parameter-norm records of section 7 five of seven have t_{acf} between 161 and 858 and are clean by this test; the asymmetric cases are the two short-correlation records and, more consequentially, the calibration systems of section 5, whose autocorrelation time is one to three samples, so that the reference band of section 6.4 was itself measured with 76 against 150. The effect is small at this span and the diagnostic separates the regimes of section 6 by a wide margin, but a ratio built from two differently excluded estimates is not the clean quantity section 3.4 describes, and lifting the cap is the first thing to change in any follow-up.

L Run inventory

Table 16 lists every training run the paper draws on. Two architectures appear: **T** is the one-layer transformer of Nanda et al. (2023), $d_{\mathrm{model}} = 128$, four heads, no layer normalisation, 226 816 parameters on modular addition at $p = 113$ and 228 608 on S_5, trained in double precision with AdamW at learning rate 10^{-3} and $\beta = (0.9, 0.98)$; **P** is the two-layer quadratic perceptron of Gromov (2023), 145 500 parameters at $p = 97$ and width 500, trained by full-batch gradient descent with no regularisation.

Three departures from the source configurations should be stated, since two of them are what put the runs in the regimes section 7.3 assigns them. Nanda et al. (2023) train full batch; we use mini-batches of 256, which is what makes this setting stochastic rather than deterministic. The S_5 weight decay of 0.2 and the S_5 task itself (Power et al., 2022) are outside that configuration. And the two S_5 runs step the optimiser twice per batch, reproducing a duplicated update in our own earlier released logs, so their effective learning rate and decay are about double the nominal ones.

What counts as an event. Memorisation and generalisation are the first logged step at which the training and the validation accuracy respectively reach $95\,\%$, with no persistence requirement. Training accuracy is measured on the current mini-batch and validation accuracy on a fixed 512-example subset of the validation split, so both carry a sampling error of a few tenths of a per cent, and the step is resolved only to the logging stride, which is 10 steps for the modular runs and 5 for S_5.

The extended reruns of section 7.3 instead require the threshold to hold over a sustained block, which is the stricter rule; the two differ on one run in this table and by 1 080 steps, and no claim in the paper turns on the difference. A dash means the criterion was not met inside the budget. Nothing here establishes that such a run would never generalise, and table 16 records the final validation accuracy so that the reader can see how far from the threshold each one ended.

Table 16: Training runs used in this paper. α is the training fraction, "final" the validation accuracy at the end of the budget. A dash in the generalisation column marks a run that had not generalised when its budget ended; such runs are censored observations, not established negatives. The name `s5_wd1` denotes the weight-decay arm of the S_5 pair, whose decay is 0.2; it is not a decay of 1.

run	task	arch	wd	α	budget	mem.	gen.	final	used in
Trajectory sketch attached									
`mod_wd1`	mod. add. p=113	T	1.0	0.30	20k	1 470	13 700	1.00	section 7.1
`mod_wd1_s43`	mod. add. p=113	T	1.0	0.30	20k	1 580	6 660	1.00	section 7.1
`mod_wd1_s44`	mod. add. p=113	T	1.0	0.30	20k	1 410	3 680	1.00	section 7.1
`s5_wd1`	S_5 composition	T	0.2	0.50	15k	1 280	6 735	1.00	section 7.1
`mod_wd0`	mod. add. p=113	T	0.0	0.30	20k	630	—	0.03	section 7.1
`s5_wd0`	S_5 composition	T	0.0	0.50	15k	1 015	—	0.02	section 7.1
Extended reruns, 120 000 *steps*									
`grokpos_s0`	mod. add. p=113	T	1.0	0.30	120k	1 450	5 110	1.00	section 7.3
`lowdata15_s0`	mod. add. p=113	T	1.0	0.15	120k	620	110 940	0.99	section 7.3
`lowdata15_s1`	mod. add. p=113	T	1.0	0.15	120k	630	—	0.94	section 7.3
`lowdata15_s2`	mod. add. p=113	T	1.0	0.15	120k	660	—	0.01	section 7.3
`lowdata20_s0`	mod. add. p=113	T	1.0	0.20	120k	800	39 600	1.00	section 7.3
`wd0_s0`	mod. add. p=113	T	0.0	0.30	120k	610	—	0.09	section 7.3
`wd0_s1`	mod. add. p=113	T	0.0	0.30	120k	630	—	0.07	section 7.3
`p211_wd0`	mod. add. p=211	T	0.0	0.16	200k	2 250	—	0.04	section 7.3
Perceptron, $p = 97$*, full batch, no regularisation*									
`a_add`	$n + m$	P	0	0.5	100k	9 170	11 950	1.00	section 7.3
`a_mul`	$n \cdot m$	P	0	0.5	100k	9 130	11 600	1.00	section 7.3
`x_no_grok`	$n^3 + nm^2 + m$	P	0	0.5	100k	9 760	—	0.01	section 7.3
`g_p1`	$(4n_1 + n_2^2)^3$	P	0	0.5	100k	7 410	12 560	1.00	section 7.3
`g_p1x`	$+ n_1 n_2$	P	0	0.5	100k	10 050	—	0.01	section 7.3
`g_p2`	$(2n_1 + 3n_2)^4$	P	0	0.5	100k	6 520	7 750	1.00	section 7.3
`g_p2x`	$- n_1^2$	P	0	0.5	100k	9 780	—	0.02	section 7.3
`g_p3`	$(5n_1^3 + 2n_2^4)^2$	P	0	0.5	100k	5 880	6 430	1.00	section 7.3
`g_p3x`	$- n_2$	P	0	0.5	100k	9 430	—	0.73	section 7.3

Four further runs repeat the top four perceptron rows with the trajectory sketch attached, at full batch and again with mini-batches of 512, for the measurement of Appendix H; a third pair repeats them at $p = 23$. Their memorisation and generalisation steps agree with the rows above to within one or two logged steps, which is what makes the comparison of Appendix H a comparison of noise rather than of learning.

Three notes on how the censored rows should be read. `lowdata15_s1` ends its budget at $94\,\%$ validation accuracy and still rising, so it is the clearest censored observation in the set rather than a negative. The generalisation step of `lowdata15_s0` is 110 940 under the sustained-block criterion used for the extended reruns and 109 860 under the plain first-crossing rule used elsewhere; nothing in the paper depends on the difference. And `g_p3x` reaches $73\,\%$ validation accuracy against a majority-class baseline of $1.3\,\%$ while being provably outside the class the architecture can represent, so it is the one row in this table that neither of the source papers accounts for.